



MXEN4000  
Mechatronic Engineering Research Project 1

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# Gauss-Newton Matrix

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Github Repo:  
<https://github.com/soapydoor17/satellite-orbit-deteremination>

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## 1 TODO

☒ Results

☒ Standardise G

☐ Discussion - Update to talk about standardised eigenvalues and eigenvectors

## 2 QUESTIONS

- Does the standardisation section look correct?

### 3 Objective

Using the gradient descent equations for correcting initial mean anomaly  $M_0$  and semi-major axis  $a$  from Doppler measurements, create the Gauss-Newton Matrix. For the typical quadratic loss gradient descent problem, this matrix is approximately equivalent to the 'Fisher Information Matrix', which shows how observable each parameter is from the data. This is usually done through getting the eigenvectors/values of the matrix. Compare the observability of the parameters of a single data point vs multiple data points.

### 4 Theory

The Gauss-Newton Matrix is given by:

$$G = \frac{1}{N} J^T J$$

Where  $N$  is the number of datapoints being considered and  $J$  is the Jacobian of the Doppler residual,  $\epsilon = \Delta f_D = f_{D \text{ pred}} - f_{D \text{ true}}$ , with respect to the parameters,  $\theta = [M_0, a]$ . The Jacobian is a  $N \times 2$  as such:

$$J(\theta) = \frac{\partial \epsilon}{\partial \theta} = \begin{bmatrix} \frac{\partial \epsilon_1}{\partial M_0} & \frac{\partial \epsilon_1}{\partial a} \\ \frac{\partial \epsilon_2}{\partial M_0} & \frac{\partial \epsilon_2}{\partial a} \\ \vdots & \vdots \\ \frac{\partial \epsilon_n}{\partial M_0} & \frac{\partial \epsilon_n}{\partial a} \end{bmatrix}$$

In order to calculate these partial derivatives, first the state vector of the satellite,  $\mathbf{X} = [\mathbf{r}, \mathbf{v}]$ , must first be found from the orbital parameters COE =  $[a, e, i, \Omega, \omega, \nu]$ . This is done using the COE2RV algorithm (Vallado 2022, 116). In this algorithm, the semiparameter ( $p$ ) is used instead of the semi-major axis ( $a$ ). Further, the true anomaly  $\nu$  should be calculated beforehand, using the Newton-Raphson Method to find the eccentric anomaly  $E$  (Vallado 2022, 65):

$$p = a(1 - e^2)$$

$$\nu = \arctan \left( \frac{\sin(E)\sqrt{1 - e^2}}{\cos(E) - e} \right)$$

Next, the ECI of the groundstation at each timestamp must be calculated. Kelso (1996) shows how to convert from geodetic coordinates (latitude/longitude/altitude) to ECI:

$$\mathbf{r}_{gs} = \begin{bmatrix} R_{\oplus} C \cos \varphi \cos \theta \\ R_{\oplus} C \cos \varphi \sin \theta \\ R_{\oplus} S \sin \varphi \end{bmatrix} \quad \mathbf{v}_{gs} = \omega_{\oplus} \times \mathbf{r}_{gs}$$

Where:

- $R_{\oplus}$  is Earth's equatorial radius (6,378.137 km as given by Vallado (2022))
- $\omega_{\oplus}$  is Earth's angular velocity ( $7.2921159 \times 10^{-5}$  rad/s as given by Vallado (2022))
- $\varphi$  is the geodetic latitude
- $\theta$  is the local sidereal time
- $C$  and  $S$  are the oblateness correction factors where:

$$\begin{aligned} - C &= \frac{1}{\sqrt{1 - f(2 - f) \sin^2 \varphi}}, \quad S = (1 - f)^2 C \\ - f &= \frac{1}{298.257} \text{ is the Earth flattening as defined by Vallado (2022)} \end{aligned}$$

blah blah blah Kelso 1996

bleep bleep (Kelso 1996)

Luckily, python libraries can be used to complete this calculation instead, meaning all we need to know is the geodetic coordinates of the groundstation, and the current time (see Appendix A for code on formatting timestamps):

```

1 from astropy.time import Time
2 from astropy.coordinates import EarthLocation, GCRS
3 import astropy.units as u
4
5 def groundstationECI(lat_deg, lon_deg, alt_m, t):
6     # Inputs:
7     #   lat_deg - latitude of groundstation in degrees
8     #   lon_deg - longitude of groundstation in degrees
9     #   alt_m   - altitude of groundstation in metres
10    #   t       - astropy.time.Time object (single time, or
11                  array of times)
12    # Outputs:
13    #   r_gs - cartesian position vector(s) of groundstation (m)
14    #   v_gs - cartesian velocity vector(s) of groundstation (m/s)
15    station = EarthLocation(lat=lat_deg*u.deg, lon=lon_deg*u.deg, height=alt_m*u.m)
16
17    pos, vel = station.get_gcrs_posvel(obstime=t)
18
19    r_gs = pos.xyz.to(u.m).value

```

```

19     v_gs = vel.xyz.to(u.m/u.s).value
20     return r_gs, v_gs

```

From here,  $\boldsymbol{\rho}$  and  $\mathbf{v}_{rel}$  can be computed, and the predicted Doppler shift can be found:

$$\begin{aligned}\boldsymbol{\rho} &= \mathbf{r} - \mathbf{r}_{gs} \\ \mathbf{v}_{rel} &= \mathbf{v} - \mathbf{v}_{gs} \\ f_D &= k \hat{\boldsymbol{\rho}} \cdot (-\mathbf{v}_{rel}), \quad k = \frac{f_c}{c}\end{aligned}$$

Thus the residual is:

$$\epsilon = \Delta f_D = f_{D \text{ pred}} - f_{D \text{ true}}$$

The relationship between the Doppler shift and satellite's cartesian state can be defined as:

$$\frac{\partial f_D}{\partial \mathbf{X}} = \left[ -k \mathbf{v}_{rel}^T \left( \frac{I}{\|\boldsymbol{\rho}\|} - \frac{\boldsymbol{\rho} \boldsymbol{\rho}^T}{\|\boldsymbol{\rho}\|^3} \right), -k \hat{\boldsymbol{\rho}}^T \right]$$

And the change of the cartesian state with respect to time is:

$$\frac{d\mathbf{X}}{dt} = \left[ \mathbf{v}, -\frac{\mu \mathbf{r}}{\|\mathbf{r}\|^3} \right]^T$$

The partial derivate of the Doppler shift with respect to the initial mean anomaly can be now finalised, where:

$$\begin{aligned}\frac{\partial f_D}{\partial M_0} &= \frac{\partial f_D}{\partial \mathbf{X}} \frac{d\mathbf{X}}{dt} \left( \frac{dM}{dt} \right)^{-1} \frac{\partial M}{\partial M_0} \\ \left( \frac{dM}{dt} \right)^{-1} &= \frac{T_p}{360} = \frac{\pi}{180n}, \quad n = \sqrt{\frac{\mu}{a^3}} \\ \frac{\partial M}{\partial M_0} &= 1\end{aligned}$$

And the partial derivative of the Doppler shift with respect to the semi-major axis is:

$$\begin{aligned}\frac{\partial f_D}{\partial a} &= \frac{\partial f_D}{\partial \mathbf{X}} \frac{d\mathbf{X}}{da} \\ \frac{d\mathbf{X}}{da} &= \frac{\partial \mathbf{X}}{\partial a} \Big|_M + \frac{\partial \mathbf{X}}{\partial M} \frac{\partial M}{\partial a} \\ \frac{\partial \mathbf{X}}{\partial a} \Big|_M &= \left[ R(1 - e \cos E) \begin{bmatrix} \cos \nu \\ \sin \nu \\ 0 \end{bmatrix}, -\frac{n}{2 - 2e \cos E} R \begin{bmatrix} -\sin E \\ \sqrt{1 - e^2} \cos E \\ 0 \end{bmatrix} \right]^T \\ \frac{\partial \mathbf{X}}{\partial M} &= \frac{1}{n} \frac{d\mathbf{X}}{dt}\end{aligned}$$

$$\frac{\partial M}{\partial a} = -\frac{3}{2}\sqrt{\frac{\mu}{a^5}} t$$

With all of the partial derivatives found for each data point, the Jacobian matrix can be formed and the Gauss-Newton matrix can be computed. Finding the eigenvalues and eigenvectors of the Gauss-Newton matrix will show how observable each parameter is to the Doppler residual.

See Appendix A for the Python code of the above math broken down into separate functions.

## 5 Method

With all over the math described in the previous section turned into python functions, the pipeline shown in Figure 1 can be used to find the Gauss-Newton Matrix from the initial TLE and following Doppler shift measurements.

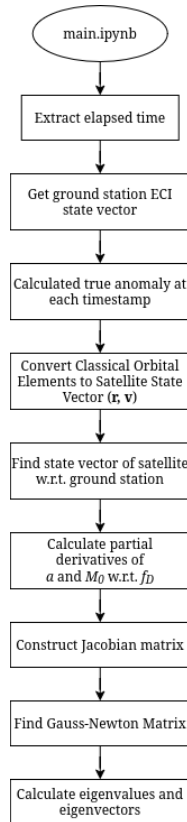


Figure 1: Block diagram of main code

### 5.1 Inputs

The two inputs required to produce Gauss-Newton matrix are an initial two line element set (TLE) and proceeding observations. For this example, the initial TLE is given as

such:

```
1 60956U 98067WW 24256.99968611 .00235683 00000+0 34489-2 0 9990
2 60956 51.6339 240.0861 0011344 359.3789 15.7195 15.54180488 2268
```

From the TLE, the following orbital parameters and datetime stamp can be found:

| Parameter                                 | Value                   |
|-------------------------------------------|-------------------------|
| Semi-major axis ( $a_{\text{true}}$ )     | 6783182.731170822 m     |
| Eccentricity ( $e_{\text{true}}$ )        | 0.0011344               |
| Inclination ( $i_{\text{true}}$ )         | 51.6339 deg             |
| RAAN ( $\Omega_{\text{true}}$ )           | 240.0861 deg            |
| Arg of perigee ( $\omega_{\text{true}}$ ) | 359.3789 deg            |
| Mean anomaly ( $M_{0 \text{ true}}$ )     | 15.7195 deg             |
| ISO Datetime stamp                        | 2024-09-12T23:59:32.880 |

Thus, the truth of the parameters in question is represented by:

$$\boldsymbol{\theta}_{\text{true}} = (M_{0 \text{ true}}, a_{\text{true}}) = (15.7195 \text{ deg}, 6783182.731170822 \text{ m})$$

The proceeding observations are imported from a `.xlsx` file, as seen in Appendix B. It provides the time of the observation (year, month, day, hour, minute, seconds as separate columns), the measured Doppler shift (Hz), and the central frequency (Hz).

The timestamp at each observation can be used with the initial orbital parameters to calculate it's respective true anomaly  $\nu$  as well as the state vector of the groundstation ( $\mathbf{X}_{gs} = [\mathbf{r}_{gs}, \mathbf{v}_{gs}]^T$ ). From there, the math previously shown can be used to get the Gauss-Newton matrix.

## 6 Results

### 6.1 Single Doppler Measurement

The single doppler measurement used in this section is:

| Field  | Value |
|--------|-------|
| Year   | 2024  |
| Month  | 9     |
| Day    | 13    |
| Hour   | 7     |
| Minute | 35    |

| Field                 | Value     |
|-----------------------|-----------|
| Second                | 5.8524243 |
| Doppler Shift (Hz)    | -9.73E+03 |
| Centre Frequency (Hz) | 4.38E+08  |

Using this as the input gave the following values

*Ground Station State Vectors*

$$\mathbf{r}_{gs} = \begin{bmatrix} -4022202.60251025 \\ -3632551.277273 \\ -3351356.54150797 \end{bmatrix} \text{ m} \quad \mathbf{v}_{gs} = \begin{bmatrix} 264.87946656 \\ -292.71712886 \\ -0.62329967 \end{bmatrix} \text{ m/s}$$

*True Anomaly*

$$\nu = 6.030164 \text{ rad}$$

*Satellite State Vectors*

$$\mathbf{r} = \begin{bmatrix} -3020280.32564041 \\ -6044413.75274605 \\ 500780.32767482 \end{bmatrix} \text{ m} \quad \mathbf{v} = \begin{bmatrix} 4470.73818599 \\ -1738.65978667 \\ 5990.43721675 \end{bmatrix} \text{ m/s}$$

*Jacobian Matrix*

$$J = \begin{bmatrix} 1.56007496 \times 10^1 & -4.20501806 \times 10^{-3} \end{bmatrix}$$

*Gauss-Newton Matrix*

$$G = \begin{bmatrix} 6.31288011 \times 10^2 & -2.00436196 \times 10^{-1} \\ -2.00436196 \times 10^{-1} & 6.36392078 \times 10^{-5} \end{bmatrix}$$

*Eigenvalues*

$$\lambda = \begin{bmatrix} 0.0 \\ 631.28807418 \end{bmatrix}$$

*Eigenvectors*

$$\mathbf{V} = \begin{bmatrix} -3.17503553 \times 10^{-4} & -9.99999950 \times 10^{-1} \\ -9.99999950 \times 10^{-1} & 3.17503553 \times 10^{-4} \end{bmatrix}$$

## 6.2 Multiple Doppler Measurements

All of the Doppler measurements used in this section are shown in Appendix B.

If the values from the previous section are correct, then it can be assumed that the functions work as intended, and thus only the results from the Jacobian matrix onwards will be shown.

*Jacobian Matrix*

$$J = \begin{bmatrix} 1.56007496 \times 10^1 & -4.20501806 \times 10^{-3} \\ 1.55842813 \times 10^1 & -4.19973428 \times 10^{-3} \\ 1.54993371 \times 10^1 & -4.17257670 \times 10^{-3} \\ 1.53480081 \times 10^1 & -4.12458914 \times 10^{-3} \\ 1.52995799 \times 10^1 & -4.10933696 \times 10^{-3} \\ 1.51958858 \times 10^1 & -4.07684685 \times 10^{-3} \\ 1.51439129 \times 10^1 & -4.06064734 \times 10^{-3} \\ 1.51395790 \times 10^1 & -4.05929904 \times 10^{-3} \\ 1.51143425 \times 10^1 & -4.05145554 \times 10^{-3} \\ 1.51101183 \times 10^1 & -4.05014396 \times 10^{-3} \end{bmatrix}$$

*Gauss-Newton Matrix*

$$G = \begin{bmatrix} 2.34234008 \times 10^2 & -6.29233512 \times 10^{-2} \\ -6.29233512 \times 10^{-2} & 1.69034631 \times 10^{-5} \end{bmatrix}$$

*Eigenvalues*

$$\lambda = \begin{bmatrix} 7.59107722 \times 10^{-11} \\ 2.34234025 \times 10^2 \end{bmatrix}$$

*Eigenvectors*

$$\mathbf{V} = \begin{bmatrix} -2.68634557 \times 10^{-4} & -9.99999964 \times 10^{-1} \\ -9.99999964 \times 10^{-1} & 2.68634557 \times 10^{-4} \end{bmatrix}$$

### 6.3 Standardisation

Eigenvalues and eigenvectors will only be meaningful if the values of  $G$  are in the same units. At the moment, each value in  $G$  currently has different units, as a result of the nature of the Jacobian matrix:

$$G = J^T J = \begin{bmatrix} \frac{\text{Hz}}{\text{deg}} \\ \frac{\text{Hz}}{\text{m}} \end{bmatrix} \begin{bmatrix} \frac{\text{Hz}}{\text{deg}} & \frac{\text{Hz}}{\text{m}} \end{bmatrix} = \begin{bmatrix} \frac{\text{Hz}^2}{\text{deg}^2} & \frac{\text{Hz}^2}{\text{deg m}} \\ \frac{\text{Hz}^2}{\text{deg m}} & \frac{\text{Hz}^2}{\text{m}^2} \end{bmatrix}$$

Thus,  $G$  needs to be scaled before analysing the eigenvalues and eigenvectors so that the directions are in comparable units. This is done with a diagonal scaling matrix  $D$ :

$$D = \text{diag}(d_{M_0}, d_a) = \begin{bmatrix} d_{M_0} & 0 \\ 0 & d_a \end{bmatrix}$$

And the scaled Gauss-Newton matrix  $\tilde{G}$  is:

$$\tilde{G} = D^T G D$$

By making  $d_{M_0} = \frac{1}{\sqrt{G_{11}}}$  and  $d_a = \frac{1}{\sqrt{G_{22}}}$ ,  $\tilde{G}_{11}$  and  $\tilde{G}_{22}$  will be 1, making the affect of  $M_0$  and  $a$  in the same scale and now comparable to each other:

$$\tilde{G} = \begin{bmatrix} 1 & \frac{G_{12}}{\sqrt{G_{11}G_{22}}} \\ \frac{G_{21}}{\sqrt{G_{11}G_{22}}} & 1 \end{bmatrix}$$

The off-diagonal term,  $\tilde{G}_{12} = \tilde{G}_{21}$ , is the correlation between the sensitivities of  $M_0$  and  $a$ .

New eigenvalues and eigenvectors can be calculated from the standardised Gauss-Newton matrix.

### 6.3.1 Single Doppler Measurement

*Gauss-Newton Matrix*

$$\tilde{G} = \begin{bmatrix} 1.0 & -1.0 \\ -1.0 & 1.0 \end{bmatrix}$$

*Eigenvalues*

$$\tilde{\lambda} = \begin{bmatrix} 0.0 \\ 2.0 \end{bmatrix}$$

*Eigenvectors*

$$\tilde{\mathbf{V}} = \begin{bmatrix} -0.70710678 & -0.70710678 \\ -0.70710678 & 0.70710678 \end{bmatrix}$$

### 6.3.2 Multiple Doppler Measurements

*Gauss-Newton Matrix*

$$\tilde{G} = \begin{bmatrix} 1.0 & -0.99913623 \\ -0.99913623 & 1.0 \end{bmatrix}$$

*Eigenvalues*

$$\tilde{\lambda} = \begin{bmatrix} 8.63771042 \times 10^{-4} \\ 1.99913623 \end{bmatrix}$$

*Eigenvectors*

$$\tilde{\mathbf{V}} = \begin{bmatrix} -0.70710678 & -0.70710678 \\ -0.70710678 & 0.70710678 \end{bmatrix}$$

## 7 Discussion

### NEEDS TO BE REWRITTEN

The Doppler model,  $f_D(M_0, a)$ , is a complicated, nonlinear relationship involving Kepler's equations, rotations and the ground station geometry. It has been linearised by replacing the model with the tangent (a first-order Taylor expansion) near a chosen point ( $\theta_{\text{true}} = (M_{0 \text{ true}}, a_{\text{true}})$ ):

$$f_D(\theta_{\text{true}} + \delta) \approx f_D(\theta_{\text{true}}) + J\delta$$

As the linearisation is done at the point of truth, the squared error is zero, meaning  $E \approx \frac{1}{2}\delta^T G \delta$ , so  $G$  describes the shape of a quadratic bowl around the true parameters and is an approximation of the Fisher Information Matrix, which shows how observable each combination of parameters are from the data. The greater the change in the loss is, the more information that direction carries. The eigenvectors are the directions in the parameter space, where each column is a different combination of the parameters ( $M_0$  and  $a$ ). The eigenvalues are the steepness along each axis, where a larger eigenvalue of a direction shows that changing it greatly affects the predicted Doppler shift, and a near-zero eigenvalue means the predicted Doppler shift barely changes along that direction, so the data cannot distinguish parameter sets that differ only along it.

For the single measurement, the eigenvalues show us that there is one direction in the parameter space that greatly informs the Doppler shift prediction ( $\lambda_2 \approx 1522.16$ ), and one direction that does not ( $\lambda_1 = 0.0$ ). This makes sense as one measurement cannot independently determine two parameters. The eigenvector associated with  $\lambda_1$  and  $\lambda_2$  respectively are:

$$\mathbf{V}_1 \approx \begin{bmatrix} 0 \\ -1 \end{bmatrix} \quad \mathbf{V}_2 \approx \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Thus, the eigenvector of  $\lambda_1$  points almost entirely in the direction of  $a$ , and  $\lambda_2$ 's eigenvector mostly points in the direction of  $M_0$ . For a single Doppler shift measurement, the observation provides almost no independent information on  $a$ , but is strongly sensitive to the value of  $M_0$ .

Using multiple observations, we see  $\lambda_1$  shifted a little bit, but still near-zero, and  $\lambda_2 \approx 749.31$ . Their respective eigenvectors are:

$$\mathbf{V}_1 \approx \begin{bmatrix} 0 \\ -1 \end{bmatrix} \quad \mathbf{V}_2 \approx \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

This, again, shows that  $a$  doesn't greatly affect the predicted Doppler shift, while the value of  $M_0$  can strongly be observed. This is supported by the fact almost all of the rows in the multi-observation  $J$  are approximately being proportional to each other. Despite there being more data, the ten observations also struggle to independently constrain  $a$ .

The multi-observation's struggle to observe the value of  $a$  is likely because the ten measurements span over 52s, under 1% of the roughly  $\frac{2\pi}{n} \approx 5560$ s orbital period. The satellite wouldn't have moved very far within this time frame and thus all of the information past the first row is mostly redundant and won't help form any new meaningful relationships.

## A Python Code

```
1 import numpy as np
2 import pandas as pd
3 from scipy.spatial.transform import Rotation as R
4
5 from astropy.time import Time
6 from astropy.coordinates import EarthLocation, GCRS
7 import astropy.units as u
8
9 MU_EARTH = 3.986004418*(10**14)
10 SPEED_OF_LIGHT = 299792458
11
12 #-----
13 # Anomaly Propagation:
14 #   M0 -> M(t) -> E -> nu
15 #-----
16
17 # Calculate mean motion (n)
18 def nCalc(a, mu=MU_EARTH):
19     # Returns the mean motion in rads/s
20     return np.sqrt(mu / a**3)
21
22 # Solve Kepler's equation  $M = E - e \sin(E)$  for E, via Newton-
23   Raphson
24 def ECalc(M_deg, e, tolerance=1e-8):
25     # Inputs:
26     #   M - Mean anomaly (degrees), scalar or array-like
27     #   e - Eccentricity
28     #   tolerance - default to 1e-8
29     # Outputs:
30     #   E - Eccentric anomaly
31
32     M_deg_arr = np.atleast_1d(np.asarray(M_deg, dtype=float))
33     M_rad = np.deg2rad(M_deg_arr)
34
35     # Initial guess
36     E = np.where(((M_rad > -np.pi) & (M_rad < 0)) | (M_rad > np
37         .pi),
38                 M_rad - e, M_rad + e)
39
40     for i in range(len(E)):
41         E_i = E[i]
42         while True:
43             sin_E = np.sin(E_i)
44             cos_E = np.cos(E_i)
45             next_E = E_i + (M_rad[i] - E_i + e * sin_E) / (1 -
46                 e * cos_E)
47             abs_diff = abs(next_E - E_i)
48             E_i = next_E
49             if abs_diff < tolerance:
```

```

47         break
48     E[i] = E_i
49
50     return E[0] if E.size == 1 else E
51
52 # Propagate mean anomaly from a reference epoch to one or more
    observation
53 #     times (given as ISO 8601 strings), then solve for eccentric
    anomaly
54 def propagateM02E(elapsed_sec, M0_deg, n, e, tolerance=1e-8):
55     # Inputs:
56     #     elapsed_sec - Elapsed time since reference epoch t0 (s)
    , i.e. (t - t0).sec
57     #     M0_deg - initial mean anomaly (deg)
58     #     n - mean motion (rads/s)
59     #     e - eccentricity
60     # Outputs:
61     #     E - eccentric anomaly (rad)
62
63     n_deg_s = np.rad2deg(n)
64     M_deg = M0_deg + n_deg_s * elapsed_sec
65
66     return ECalc(M_deg, e, tolerance=tolerance)
67
68 # Calculate true anomaly (nu)
69 def nuCalc(E, e):
70     # Inputs:
71     #     E - Eccentric anomaly (radians)
72     #     e - Eccentricity
73     # Outputs:
74     #     nu - True Anomaly (radians)
75     nu = np.arctan2(np.sin(E) * np.sqrt(1-e**2), np.cos(E) - e)
76     return nu
77
78
79 #-----
80 # COE2RV:
81 #     Convert Classical Orbital Elements (COE) to
82 #     Cartesian Position and Velocity vectors (RV)
83 #-----
84 def COE2RV(coe, mu=MU_EARTH):
85     # INPUT:
86     #     coe is an array of the Keplerian Orbital Elements
87     #     a - semi-major axis (m)
88     #     e - eccentricity
89     #     i - inclination (deg)
90     #     node - right ascension of the ascending node (deg)
91     #     arg - argument of perigee (deg)
92     #     nu - true anomaly (radian)
93     #     mu - gravitational parameters (=GM). Default set to the
    value for Earth
94

```

```

95     # OUTPUT:
96     #   r - position vector of satellite
97     #   v - velocity vector of satellite
98
99
100    a, e, i, node, arg, nu = coe
101
102    sin_nu = np.sin(nu)
103    cos_nu = np.cos(nu)
104
105    # Calculate semiparameter (p)
106    p = a * (1-e**2)
107
108    # Perifocal Coordinate System
109    r_PQW = np.zeros(3)
110    v_PQW = np.zeros(3)
111
112    r_PQW[0] = (p * cos_nu) / (1 + e*cos_nu)
113    r_PQW[1] = (p * sin_nu) / (1 + e*cos_nu)
114
115    v_PQW[0] = -1 * np.sqrt(mu/p) * sin_nu
116    v_PQW[1] = np.sqrt(mu/p) * (e + cos_nu)
117
118    # Rotation Matrix
119    R_node_z = R.from_euler('z', np.deg2rad(node)).as_matrix()
120    R_i_x     = R.from_euler('x', np.deg2rad(i)).as_matrix()
121    R_arg_z   = R.from_euler('z', np.deg2rad(arg)).as_matrix()
122    R_total   = R_node_z @ R_i_x @ R_arg_z
123
124    # Rotate Perifocal to IJK
125    r_IJK = R_total @ r_PQW
126    v_IJK = R_total @ v_PQW
127
128    return r_IJK, v_IJK, R_total
129
130    #-----
131    # Time conversion
132    #-----
133    def ISOString(year, month, day, hour, minute, second):
134        # Inputs:
135        #   year, month, day, hour, minute - integer
136        #   second - float
137        # Output:
138        #   dt_str - ISO 8601 datetime string
139        dt_str = (
140            f"{int(year):04d}-{int(month):02d}-{int(day):02d}T"
141            f"{int(hour):02d}:{int(minute):02d}:{second:09.6f}"
142        )
143        return dt_str
144
145    # Get astropy.time.Time object of a specific row from a
146    DataFrame

```

```

146 # Apply to whole dataframe:
147 #     df["Time"] = df.apply(getTime, axis=1)
148 def getTime(row, t_scale='utc'):
149     # Input:  row of DataFrame that has the columns: "Year", "
150             Month", "Day", "Hour", "Minute", "Second"
151     #         t_scale - time scale, default 'utc'
152     # Output: astropy.time.Time object for the inputted row
153     iso_str = ISOString(
154         row["Year"], row["Month"], row["Day"],
155         row["Hour"], row["Minute"], row["Second"]
156     )
157     return Time(iso_str, scale=t_scale)
158
159 #-----
160 # Groundstation Vectors:
161 #   Geodetic coords -> ECI
162 #-----
163 def groundstationECI(lat_deg, lon_deg, alt_m, t):
164     # Inputs:
165     #   lat_deg - latitude of groundstation in degrees
166     #   lon_deg - longitude of groundstation in degrees
167     #   alt_m   - altitude of groundstation in metres
168     #   t       - astropy.time.Time object (single time, or
169                 array of times)
170     # Outputs:
171     #   r_gs - cartesian position vector(s) of groundstation (m
172             )
173     #   v_gs - cartesian velocity vector(s) of groundstation (m
174             /s)
175     station = EarthLocation(lat=lat_deg*u.deg, lon=lon_deg*u.
176                             deg, height=alt_m*u.m)
177
178     pos, vel = station.get_gcrs_posvel(obstime=t)
179
180     r_gs = pos.xyz.to(u.m).value
181     v_gs = vel.xyz.to(u.m/u.s).value
182     return r_gs, v_gs
183
184 #-----
185 # Doppler Shift Calculations from r and v
186 #-----
187 # rho - position vector of the satellite w.r.t groundstation
188 def rhoCalc(r, r_gs):
189     return r-r_gs
190
191 # rho_hat - unit vector of rho
192 def rho_hatCalc(rho):
193     return rho / np.sqrt(rho.dot(rho))
194
195 # v_rel - velocity vector of the satellite w.r.t groundstation

```

```

193 def v_relCalc(v, v_gs):
194     return v-v_gs
195
196 # frequency-to-range-rate scale factor
197 def kCalc(f_c, c=SPEED_OF_LIGHT):
198     return f_c/c
199
200
201 # Doppler shift calculation
202 # NOTE: NOT USED?
203 def fDCalc(k, rho_hat, v_rel):
204     return k * rho_hat * (-v_rel)
205
206 # Residual Calculation
207 # NOTE: NOT USED?
208 def residual(y_pred, y_true):
209     return y_pred-y_true
210
211 #-----
212 # Gradient Calculations
213 #-----
214 # Relationship between Doppler shift and satellite's cartesian
    state
215 def dfD_dXCalc(k, v_rel, rho, rho_hat):
216     mag_rho = np.sqrt(rho.dot(rho))
217
218     I3 = np.eye(3)
219
220     dfD_dr = -k * v_rel.T @ ( I3/mag_rho - (np.outer(rho, rho))
        /(mag_rho**3) )
221     dfD_dv = -k * rho_hat.T
222     return np.concatenate([dfD_dr, dfD_dv])
223
224
225 # Derivative of the satellite's cartesian state
226 def dX_dtCalc(v, r, mu=MU_EARTH):
227     mag_r = np.sqrt(r.dot(r))
228     dv_dt = -mu * r / (mag_r**3)
229
230     return np.concatenate([v, dv_dt])
231
232 # Initial Mean Anomaly Gradient
233 def dfD_dM0Calc(dfD_dX, dX_dt, n):
234     # Inputs:
235     #   dfD_dX - Change in Doppler measurement w.r.t.
        satellite's cartesian state (1x6 array)
236     #   dX_dt - Change in satellite's cartesian state w.r.t.
        time (6x1 array)
237     #   n - Mean motion (rad/s)
238     # Outputs:
239     #   dfD_dM0 - Change in Doppler shift w.r.t. Initial Mean
        Anomaly (Hz/deg)

```

```

240     dM_dt_reciprocal = 1 / np.rad2deg(n)
241
242
243     return dfD_dX @ dX_dt * dM_dt_reciprocal
244
245 # Semi-Major Axis Gradient
246 def dfD_daCalc(dfD_dX, dX_dt, elapsed_sec, rot_mat, n, a, e, E,
247               nu, mu=MU_EARTH):
248     # Inputs:
249     #   dfD_dX      - Change in Doppler measurement w.r.t.
250                     satellite's cartesian state (1x6 array)
251     #   dX_dt       - Change in satellite's cartesian state w.r.
252                     .t. time (6x1 array)
253     #   elapsed_sec - Elapsed time since reference epoch t0 (s)
254                     , i.e. (t - t0).sec
255     #   rot_mat     - Perifocal-to-ECI rotation matrix (from
256                     COE2RV)
257     #   n           - Mean motion (rad/s)
258     #   a           - Semi-major axis (m)
259     #   e           - Eccentricity
260     #   E           - Eccentric anomaly (rad)
261     #   nu          - True anomaly (rad)
262     #   mu          - Gravitational parameter (m3/s2) (
263                     Defaults to Earth's mu value)
264     # Outputs:
265     #   dfD_da      - Change in Doppler shift w.r.t semi-major axis
266     # NOTE: For calculations in this function, M is in radians
267
268     nu_arr = np.array([np.cos(nu), np.sin(nu), 0])
269     E_arr = np.array([-np.sin(E), np.sqrt(1-e**2) * np.cos(E),
270                      0])
271
272     dr_da_M = (1 - e*np.cos(E)) * rot_mat @ nu_arr
273     dv_da_M = -n / (2- 2*e*np.cos(E)) * rot_mat @ E_arr
274
275     dX_da_M = np.concatenate([dr_da_M, dv_da_M])
276     dX_dM = dX_dt / n
277     dM_da = -3/2 * np.sqrt(mu / (a**5)) * elapsed_sec
278
279     dX_da = dX_da_M + dX_dM * dM_da
280
281     return dfD_dX @ dX_da
282
283 #-----
284 # Jacobian Calculations
285 #-----
286 def Jacobian(dfD_dM0, dfD_da):
287     # Create Jacobian Matrix
288     # Inputs:
289     #   dfD_dM0 - Initial mean anomaly gradient (scalar or Nx1
290             array)
291     #   dfD_da  - Semi-major axis gradient (scalar or Nx1

```

```

    array)
284 # Outputs:
285 #   J - Outputs Nx2 array, where N is the number of
    observations
286
287 # Convert scalar values to 1D arrays
288 dfD_dM0 = np.atleast_1d(dfD_dM0)
289 dfD_da = np.atleast_1d(dfD_da)
290
291 N = len(dfD_dM0)          # Number of entries
292
293 # Create matrix
294 J = np.zeros((N, 2))
295 for i in range(N):
296     J[i] = np.array([dfD_dM0[i], dfD_da[i]])
297
298 return J
299
300 #-----
301 # Gauss-Newton Matrix
302 #-----
303 def GaussNewton(J):
304     # Create Gauss-Newton Matrix
305     # Inputs:
306     #   J - Nx2 Jacobian Matrix, where N is the number of
    observations
307     # Outputs:
308     #   G - 2x2 Gauss-Newton Matrix
309
310     N = J.shape[0]
311     return 1/N * J.T @ J
312
313 def standardisedGaussNewton_M0_a(J):
314     # Create Standardised Gauss-Newton Matrix
315     # Inputs:
316     #   J - Nx2 Jacobian Matrix, where N is the number of
    observations
317     # Outputs:
318     #   G_tilde - 2x2 Gauss-Newton Matrix
319
320     G = GaussNewton(J)
321     sigma_M0 = 1 / np.sqrt(G[0, 0])
322     sigma_a = 1 / np.sqrt(G[1, 1])
323     D = np.array([[sigma_M0, 0], [0, sigma_a]])
324
325     return D.T @ G @ D

```

## B Fake Doppler Measurement Data

| Year | Month | Day | Hour | Minute | Second      | Doppler Shift (Hz) | Centre Frequency (Hz) |
|------|-------|-----|------|--------|-------------|--------------------|-----------------------|
| 2024 | 9     | 13  | 7    | 35     | 5.8524243   | -9.73E+03          | 4.38E+08              |
| 2024 | 9     | 13  | 7    | 35     | 7.619208978 | -9.73E+03          | 4.38E+08              |
| 2024 | 9     | 13  | 7    | 35     | 16.70989313 | -9.75E+03          | 4.38E+08              |
| 2024 | 9     | 13  | 7    | 35     | 32.81289115 | -9.77E+03          | 4.38E+08              |
| 2024 | 9     | 13  | 7    | 35     | 37.94155477 | -9.78E+03          | 4.38E+08              |
| 2024 | 9     | 13  | 7    | 35     | 48.88342118 | -9.79E+03          | 4.38E+08              |
| 2024 | 9     | 13  | 7    | 35     | 54.34751622 | -9.79E+03          | 4.38E+08              |
| 2024 | 9     | 13  | 7    | 35     | 54.80255137 | -9.79E+03          | 4.38E+08              |
| 2024 | 9     | 13  | 7    | 35     | 57.45041013 | -9.80E+03          | 4.38E+08              |
| 2024 | 9     | 13  | 7    | 35     | 57.89331211 | -9.80E+03          | 4.38E+08              |

## References

- Kelso, T.S. 1996. “Orbital Coordinate Systems, Part III.” CelesTrak, January. Accessed September 27, 2026. <https://celestrak.org/columns/v02n03/>.
- Martens, James. 2020. “New Insights and Perspectives on the Natural Gradient Method.” Pre-published, September 19, 2020. Accessed September 22, 2026. <https://doi.org/10.48550/arXiv.1412.1193>. <http://arxiv.org/abs/1412.1193>.
- Vallado, David A. 2022. *Fundamentals of Astrodynamics and Applications*. Fifth edition, first printing. Space Technology Library 21. Torrance, CA: Microcosm Press.